

# Report 27 - QuasiSteadyPipe Model

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## Abstract

This report documents the QuasiSteadyPipe model family in OpenHPLjl. Catalogue status: **prototype**.

## 1 Updated Engineering Context

Algebraic conduit without water inertia. It belongs to the Rigid pipes and waterways domain.

## 2 Generalized Concept Model

The model preserves the domain connector interface while allowing internal fidelity to evolve independently.

## 3 Alternative Models Available in Literature

This family sits on the common fidelity ladder: boundary/algebraic, lumped dynamic, nonlinear physics-based, and distributed/high-order.

## 4 Governing Mathematics

$$H_u - H_d = RQ|Q|$$

The production model may add states, constitutive relations, lookup maps, or spatial discretization.

## 5 OpenHPL / Modelica Interpretation

Report 9 maps this family to the closest OpenHPL, Modelica Standard Library, or OpenIPSL analogue.

## 6 Julia / ModelingToolkit Implementation

Source: `src/Waterways/RigidPipes/QuasiSteadyPipe.jl`. The file is either an implemented component or an explicit Julia source specification.

## 7 Component Connections and Assembly

The model connects through the stable physical connector for its domain and should not expose internal fidelity choices to neighbouring components.

## 8 Numerical Experiment / Validation

Promotion requires an equation-level test, component experiment, limiting-case comparison, structural check, and connected-system test.

## 9 Expected Outputs

Expected outputs are the physical states, connector variables, and conservation or residual quantities needed to verify this family.

## 10 Engineering Interpretation

Use as a limiting case and initialization aid.

### Conclusion

QuasiSteadyPipe now has an explicit source/report pair. Promotion to baseline requires validated equations, parameters, tests, and reference behaviour.